T5 Dual-sided page format

Validate fixed V2 rails, wrap-first layer changes and a working circuit on both faces.

V2 format test

V+ · top rail

V+ A

H1 · no net

A A

BUS A · above centre ring

continue tape to edge and fold

H2 · no net

BUS B · below centre ring

B A

R2 · 1 kΩ

Side A load

R2 and LED2 run from BUS B down to GND. Test 2 drives BUS B from side B, through the outer-edge wrap.

Routing rule

Wrap around an edge to change faces. A soldered wire via is an exception only when no safe wrap path exists.

+

LED2

H3 · no net

GND · wide bottom rail

GND A

Print at 100 %: this bar should measure 100 mm.

T5 Dual-sided format: side B

This side prints behind sheet 1. The four rails meet side A only at their edge wraps.

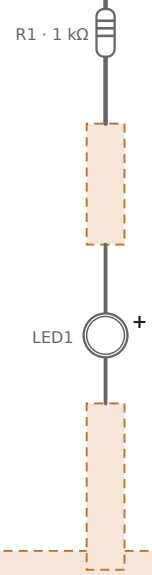
V+ · top rail

V+ B

H1 · no net

Side B load

R1 and LED1 run from V+ down to BUS A. Test 1 returns BUS A to GND through the opposite face.



continue tape to edge and fold

BUS A · above centre ring

A B

H2 · no net

BUS B · below centre ring

B B

No page vias

V+ wraps over the top, GND over the bottom, and both buses around the outer edge. Build no through-page via on this test.

H3 · no net

GND · wide bottom rail

GND B

Print at 100 %: this bar should measure 100 mm.

T5 Build and test the dual-sided format

Print sheets 1 and 2 double-sided on card, flipping on the long edge. Print sheets 3 and 4 once.

H1 · no net

You need

- One card printed double-sided (sheets 1 and 2), a three-hole punch and a three-ring binder
- 5 mm copper tape for V+ and the buses; 12 mm copper tape for GND
- Two 1 kΩ resistors, R1 and R2; two red 5 mm LEDs, LED1 and LED2
- A soldering iron and solder, clip leads and a multimeter
- A fused 3 × AA holder, 4.5 V nominal

1 · Print and inspect

1. Check the 100 mm scale bar on both faces. Hold the card to a light: each side-B rail must sit directly behind its side-A partner.

2. Punch the three mechanical holes. Confirm that no copper guide enters an 8 mm hole keep-out.

3. Mark the two faces A and B before laying tape.

2 · Lay the four rails

1. Lay V+ as one continuous 5 mm strip from the side-A spine, across the top rail, over the top edge and along the side-B top rail to its spine.

2. Lay GND the same way with 12 mm tape, using the bottom-edge wrap.

3. Lay BUS A as one continuous 5 mm strip from the side-A spine, around the outer edge and back to the side-B spine. Repeat for BUS B.

4. Burnish every fold. Do not add a through-page via.

5. Follow each solid arrow to the physical edge even though the printed lane stops short. Round circle-and-dot marks are electrical test or clip points, not vias.

H2 · no net

3 · Add the loads

1. On side B, add R1 and LED1 between V+ and BUS A. Put the LED's long anode leg toward V+.

2. On side A, add R2 and LED2 between BUS B and GND. Put the LED's long anode leg toward BUS B.

3. Solder every component lead to its tape lane. Check that no trimmed lead reaches the other face.

4 · Continuity before power

1. Measure each rail from its side-A test point to its side-B test point. Record the resistance on sheet 4.

2. V+, BUS A and BUS B must each be 0.1 Ω or less. The wide GND rail must be 0.05 Ω or less.

3. Check every pair of different rails. Every reading must be OL; find any bridge before connecting power.

5 · Powered proof

1. Connect the fused holder red lead to V+ A and black lead to GND A.

2. **Test 1:** clip BUS A A to GND B. LED1 on side B must light. This path uses the V+ top wrap, BUS A outer wrap and GND bottom wrap.

3. Remove that clip. **Test 2:** clip V+ B to BUS B B. LED2 on side A must light. This path uses the V+ top wrap and BUS B outer wrap.

4. Disconnect power before moving or inspecting tape.

6 · Flex and inspect

1. Flex each wrapped edge through 100 cycles, measure all four rails, and repeat at 1,000 cycles.

2. Repeat both powered tests while gently opening and closing the card in the binder. Record flicker or resets.

3. Inspect for cracked copper, lifted adhesive, torn card or a wrap that has crept sideways.

Pass or fail

- **Pass:** both LEDs work only in their stated test, all four rails meet their resistance limits, no rail shorts another and no flicker occurs through 1,000 cycles.
- **Fail:** any wrap needs a through-page via to complete this layout, or any copper, paper or electrical failure appears.
- If a future lesson cannot reach an edge safely, record why before permitting a soldered wire via as a documented exception.

H3 · no net

T5 Results

Record wrap resistance in ohms and decide whether this becomes the V2 page format.

Side A to side B continuity

Measure at the labelled test points with power disconnected.

	New	100	1,000
V+			
BUS A			
BUS B			
GND			

Powered proof

	Before flex	After flex
LED1 / BUS A		
LED2 / BUS B		

Format decision

V+ stayed above the first hole	
GND stayed at the bottom	
BUS A and BUS B cleared the centre hole	
Side A/B registration was usable	
All four edge wraps passed	
Any page via was required	
Adopt this format for the first V2 lesson	

Notes

Page-via exception and reason	
Damage, flicker or routing changes	
Builder / date	

Print at 100 %: this bar should measure 100 mm.